

Paper Title

Author Name

Abstract—

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I. INTRODUCTION

Deep learning has achieved strong performance in many radar applications, including target detection, classification, and recognition. Its performance, however, depends strongly on the quantity, diversity, and annotation quality of the available data. This dependence is particularly restrictive for maritime radar because existing measured datasets cover only a limited set of sea states, radar parameters, viewing geometries, and target conditions. A model trained on such data may not generalize to an unobserved wind speed, wave direction, grazing angle, or clutter regime. Expanding a measured dataset is also difficult: sea experiments require suitable platforms, radar equipment, environmental sensors, synchronized observations, and favorable weather, while a desired sea condition cannot be reproduced on demand. The shortage and limited diversity of measured radar data have therefore become a major constraint on the development of data-driven maritime sensing methods.

Numerical simulation provides a practical way to supplement measured data because environmental and radar parameters can be controlled independently and a large number of labeled samples can be generated repeatedly. For radar applications, however, computational efficiency alone is insufficient. A simulated sea surface must preserve the statistical roughness of the background sea, respond reasonably to different wind and wave conditions, and represent the local structures that produce strong or non-Gaussian radar returns. High-fidelity hydrodynamic methods can resolve nonlinear wave evolution and breaking interfaces in detail, while full-wave electromagnetic methods can describe the associated diffraction and multipath interactions. Their volume-mesh evolution, fine spatial discretization, or global electromagnetic interactions nevertheless impose substantial computational and storage costs, making repeated simulation over many environmental and radar conditions difficult. Spectral sea-surface and facet-based scattering models are considerably more efficient, but a conventional linear realization does not explicitly represent strongly nonlinear or overturning crests. A useful data-generation method must therefore balance physical credibility, environmental adaptability, computational cost, and compatibility with radar scattering calculations.

Sea-surface roughness is inherently multiscale. Long gravity waves determine the large-scale tilt and motion of the surface, whereas shorter wind waves modify local slopes and radar-resonant components. Consequently, the observed return depends jointly on sea state, incidence angle, radar wavelength, and surface geometry. High-resolution measurements show that sea clutter may exhibit pronounced non-Gaussian tails

and intermittent sea spikes, which can be confused with target responses [?], [?], [?], [?], [?]. Directional wave spectra, such as the unified Elfouhaily spectrum, provide an efficient description of the gravity and gravity-capillary energy distribution [?]. Nevertheless, a linear spectral realization is approximately Gaussian and mainly preserves second-order statistics. Nonlinear hydrodynamic interactions sharpen crests, smooth troughs, and change the distributions of slopes and curvatures, thereby influencing the scattering coefficient and Doppler spectrum [?]. The sea-surface skewness function and its small-scale-wave parameter also affect the upwind-downwind asymmetry of radar backscattering [?], while convolutions of short-wave spectral components can contribute appreciably to the normalized backscattering cross section under different radar frequencies, wind speeds, and incidence angles [?]. Thus, a radar-oriented sea model should represent not only the total spectral energy but also nonlinear crest geometry and the directional organization of shorter wind waves.

The return from an ordinary rough sea surface is commonly described by quasi-specular, Bragg, or two-scale scattering formulations. Their relative contributions vary with incidence angle, and both quasi-specular and Bragg mechanisms may be required in the transition range [?]. Radar observations of marine film slicks further suggest that the source of short wind waves can depend on propagation direction and cannot always be represented by conventional wind-input terms alone [?]. For steep and breaking waves, additional scattering mechanisms become important. Numerical and experimental studies have related low-grazing-angle sea spikes to steepening or breaking crests [?], [?], [?]. Full-wave and hybrid models have investigated crest diffraction, rough-face scattering, and multipath interactions [?], [?], [?], [?], [?]. Facet-based methods have introduced breaking-wave contributions into larger sea surfaces and analyzed their effects on normalized radar cross section (NRCS), sea spikes, and synthetic aperture radar images [?]. Three-dimensional plunging-breaker models combined with ray tracing have also shown that a breaker crest can generate target-like peaks in radar cross section and high-resolution range profiles [?].

Existing approaches present two complementary limitations. On the electromagnetic side, rigorous or high-order numerical methods can resolve diffraction, rough-face scattering, and multipath interactions from breaking waves, but they require fine discretization and repeated solution of large scattering systems [?], [?]. At X-band, conventional numerical discretization may require surface samples much smaller than the radar wavelength, making time-evolving and large-area calculations particularly expensive [?]; even accelerated surface-scattering methods report substantial costs for multiview or full-polarimetric calculations [?]. Lower-cost electromagnetic

approximations also have restricted validity. The traditional two-scale model depends on a prescribed separation between quasi-specular and Bragg regimes and loses accuracy where the two mechanisms coexist [?], whereas steep or multivalued breaker faces introduce crest diffraction, shadowing, and multipath contributions that are not represented by an ordinary rough-surface term alone [?], [?], [?]. Thus, electromagnetic fidelity and computational efficiency remain difficult to obtain simultaneously for large-scale radar-data generation.

On the sea-surface side, computationally efficient spectral and replacement-based methods have different limitations. Linear spectral synthesis preserves the prescribed second-order spectrum but cannot reproduce crest–trough asymmetry or overturning geometry, while Creamer- or Lie-type nonlinear surfaces improve slopes and higher-order statistics but generally remain single-valued height fields [?], [?]. Existing breaking-wave studies often use a selected LONGTANK stage, a measured breaker profile, or a prescribed analytical three-dimensional breaker and then insert or superimpose that geometry into the scattering scene [?], [?], [?], [?]. These constructions are effective for investigating the radar signature of a particular breaker, but their emphasis is not the automatic generation of diverse surfaces conditioned jointly on wind speed, wind direction, short-wave organization, and local crest state [?], [?]. This limitation is important because radar observations indicate that short-wave generation and damping can depend strongly on propagation direction and surface condition [?]. Directly superimposing an independently generated short-wave field without band separation or renormalization can duplicate spectral contributions already contained in the background surface [?], [?], while replacing a region with a prescribed breaker profile may limit geometric continuity and adaptability across sea states [?], [?], [?]. A rapid dataset-oriented model should therefore connect a statistically constrained background, directionally organized short waves, and locally triggered curling geometry within the same environmental parameterization. Its advantage should be evaluated jointly in terms of physical representativeness, environmental flexibility, radar applicability, and measured computational scalability.

This letter proposes a hierarchical sea-surface simulation method intended for rapid radar-data generation under variable environmental conditions. First, an Elfouhaily directional spectrum and a wind-dependent nonlinear Lie transformation are used to construct a statistically constrained nonlinear background surface. Second, a selected short-wave band is replaced by an equal-energy directional component, forming shorter wind-wave ridges with a prescribed dominant propagation direction without repeatedly injecting spectral energy. Third, candidate crests are identified from elevation, propagation-direction slope, and curvature, after which a compactly supported coordinate transformation generates a localized forward-extending and downward-curling breaker. Wind speed, wind direction, directional spreading, short-wave scale, and local curling parameters can be varied to produce different sea-surface realizations, while the facet representation allows the generated surfaces to be used directly in radar scattering and echo simulations. Because the nonlinear and

curling operations are applied through spectral and sparse local surface updates, the method avoids repeated three-dimensional fluid evolution and dense full-wave interaction for every generated realization. It is therefore designed to retain radar-relevant breaker geometry while remaining suitable for large, controllable synthetic radar datasets.

The method is evaluated from surface geometry to radar observables and computational scalability. The background and short-wave-enriched surfaces are first examined using spectral consistency and along-wind and crosswind mean-square slopes under different wind conditions. The localized breaker is then assessed using crest geometry, facet-orientation statistics, radar-facing projected area, and paired unit-footprint scattering enhancement. Radar echoes generated from the complete surface are evaluated through range–time intensity, temporal correlation, and amplitude-distribution statistics. Finally, the measured runtime is decomposed into surface generation, facet-level scattering, echo formation, and input/output costs; its scaling with surface resolution and dataset size is compared with reported X-band numerical methods, with differences in output scope stated explicitly [?], [?], [?]. Section II presents the hierarchical surface model. Section III describes the validation metrics and radar echo generation. Section IV reports the surface and radar comparisons. Section V analyzes computational cost and dataset scalability, and Section VI concludes this letter.

A. Overall Framework

The objective of the proposed framework is to generate physically interpretable sea-surface realizations under controllable environmental conditions while retaining a computational form suitable for radar echo simulation. The environmental parameter vector is denoted by

$$\mathbf{p}_e = \{U_{10}, \phi_w, \Omega, H_s, L_x, L_y, \Delta x, \Delta y\}, \quad (1)$$

where U_{10} is the wind speed at a height of 10 m, ϕ_w is the wind direction, Ω denotes the inverse wave age, H_s is the prescribed or normalized significant wave height, L_x and L_y define the horizontal domain, and Δx and Δy are the sampling intervals. A random variable set ξ controls the spectral phases and therefore produces independent realizations under the same environmental condition. The radar parameter vector is written as

$$\mathbf{p}_r = \{f_c, \theta_i, \phi_r, \mathcal{P}, B, N_p\}, \quad (2)$$

where f_c , θ_i , ϕ_r , $\mathcal{P}$, B , and N_p denote the carrier frequency, incidence angle, radar azimuth, polarization, signal bandwidth, and pulse number, respectively. Separating $\mathbf{p}_e$, $\mathbf{p}_r$, and ξ allows the sea state, radar observation, and random realization to be varied independently during dataset generation.

The surface construction contains three successive levels. First, a linear random surface is synthesized from the Elfouhaily directional spectrum [?]. Let $S_E(\mathbf{k}; \mathbf{p}_e)$ be the two-dimensional spectrum and $\varphi_{\mathbf{k}}(\xi)$ be the random phase asso-

Placeholder for the overall framework figure

Left to right: environmental parameters and random phases $\rightarrow$ linear Elfouhaily surface $\rightarrow$ nonlinear Lie background $\rightarrow$ equal-energy directional short-wave reconstruction $\rightarrow$ steep-crest detection and localized curling transformation $\rightarrow$ triangular facets and radar echo generation.

The final version should show the three surface states using the same spatial domain and mark the locally triggered curling region.

Fig. 1. Overall framework of the proposed hierarchical sea-surface and radar-echo simulation method.

ciated with wave vector $\mathbf{k}$. The linear elevation is expressed compactly as

$$\eta_L(x, y) = \mathcal{F}^{-1} \left\{ \sqrt{2S_E(\mathbf{k}; \mathbf{p}_e) \Delta k_x \Delta k_y} \times \exp[j\varphi_{\mathbf{k}}(\boldsymbol{\xi})] \right\}, \quad (3)$$

where conjugate symmetry is imposed to obtain a real-valued surface. A wind-dependent nonlinear Lie operator $\mathcal{L}$ is then applied to introduce crest sharpening and crest-trough asymmetry while constraining the large-area slope statistics:

$$\eta_{NL} = \mathcal{L}[\eta_L; U_{10}, \phi_w]. \quad (4)$$

This level provides the nonlinear background surface but retains a single-valued elevation representation.

Second, a prescribed short-wave band $\mathcal{K}_s$ is reorganized to form wind-wave ridges with a dominant propagation direction. Instead of directly adding an independent component, the original band is removed and replaced by an equal-energy directional realization:

$$\eta_D = \eta_{NL} - \mathcal{B}_s[\eta_{NL}] + \eta_s^{\text{dir}}, \quad (5)$$

subject to

$$\int_{\mathcal{K}_s} S_s^{\text{dir}}(\mathbf{k}) d\mathbf{k} = \int_{\mathcal{K}_s} S_{NL}(\mathbf{k}) d\mathbf{k}. \quad (6)$$

Here, $\mathcal{B}_s[\cdot]$ is a band-pass operator, η_s^{dir} is the reconstructed directional component, and S_s^{dir} and S_{NL} are the corresponding spectral densities. Equation (??) prevents repeated energy injection into the selected band while allowing its directional distribution to be controlled.

Third, a local crest score $Q(x, y)$ is computed from the normalized elevation, the slope along the dominant propagation direction, and the negative directional curvature. The curling center is selected as

$$(x_c, y_c) = \arg \max_{(x, y) \in \mathcal{D}_c} Q(x, y), \quad (7)$$

$$Q(x, y) = w_h Q_h + w_s Q_s + w_c Q_c,$$

where $\mathcal{D}_c$ is the admissible crest region and w_h , w_s , and w_c are weighting coefficients. A local coordinate system (u, v, z) is established at (x_c, y_c) , with u along the dominant propagation direction and v along the crest. The final curled surface is represented parametrically as

$$\mathbf{r}_C(u, v) = \mathbf{r}_D(u, v) + W(u, v) \mathbf{d}_C(u, v, z; \mathbf{p}_c), \quad (8)$$

where $\mathbf{r}_D = [x_D, y_D, \eta_D]^T$ is the directional-wave surface, $W(u, v)$ is a compact support function, $\mathbf{d}_C$ is the curling displacement, and $\mathbf{p}_c$ contains the local length, width, height, forward-extension, and curling parameters. Unlike η_L , η_{NL} , and η_D , the final geometry is not restricted to the form $z = \eta(x, y)$; the parametric coordinates in (??) permit locally overturned facets.

The parametric surface is triangulated to form the mesh $\mathcal{M}_C$, which provides facet locations, areas, and normal vectors for the radar model. The complete generation chain can therefore be summarized as

$$(\mathbf{p}_e, \boldsymbol{\xi}) \xrightarrow{\mathcal{G}_E} \eta_L \xrightarrow{\mathcal{L}} \eta_{NL} \xrightarrow{\mathcal{D}_s} \eta_D \xrightarrow{\mathcal{C}} \mathcal{M}_C \xrightarrow{\mathcal{R}(\mathbf{p}_r)} \mathbf{y}, \quad (9)$$

where $\mathcal{G}_E$, $\mathcal{D}_s$, $\mathcal{C}$, and $\mathcal{R}$ denote spectral synthesis, directional short-wave reconstruction, localized curling, and radar echo generation, respectively, and $\mathbf{y}$ is the simulated radar data. This hierarchical formulation preserves the efficiency of spectral synthesis over most of the domain and confines the additional curling computation to a sparse, physically selected region.

REFERENCES